Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	SWTID-2026-5297
Project Name	Script-Controlled ACL –Restricted Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which users access records and applications	ServiceNow UI / Service Portal
2	Application Logic-1	Checks the user's role and permissions	ServiceNow ACL
3	Application Logic-2	Evaluates the field value and applies conditional access rules	JavaScript / ACL Script
4	Application Logic-3	Determines whether the user can read, write, create or delete the record	ServiceNow Access Control
5	Database	Stores records, users, roles and field values	ServiceNow Database
6	Cloud Database	Cloud-hosted storage for application and configuration data	ServiceNow Cloud Database
7	File Storage	Stores attachments associated with records, if required	ServiceNow Attachment Storage

8	External API-1	Optional integration for retrieving external user or authorization information	REST API
9	External API-2	Optional integration with external enterprise systems	REST / SOAP API
10	Machine Learning Model	Not required; access decision is rule/script based	Not Applicable
11	Infrastructure	Hosts the application, ACL engine, database and configuration	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	JavaScript is used for implementing conditional ACL scripts and access logic	JavaScript
2	Security	Restricts record/field access based on user roles and field values. Prevents unauthorized read/write operations	ServiceNow ACL, Role-Based Access Control, Authentication, Authorization
3	Scalable Architecture	ACL rules are centrally managed and can be applied across multiple tables and applications	ServiceNow Platform Architecture
4	Availability	Cloud-based ServiceNow platform provides continuous access to the application and security rules	ServiceNow Cloud Infrastructure

5	Performance	ACL scripts perform lightweight condition checks before allowing database operations, reducing unnecessary access	Server-side JavaScript / ServiceNow ACL Engine